Evaluation Validity in SDN Intrusion Detection: Controller Observability and Held-Out-Host Generalization on InSDN

A measurement and evaluation-protocol study

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THE PROBLEM

Detectors are scored on data the controller never sees

SDN papers justify detection by the controller's global view.

They then evaluate on CSV features computed at the victim host.

Nobody checked what the controller actually receives.

InSDN also recorded its controller's OpenFlow traffic. We parsed all 44 of those captures.

What papers score on
CICFlowMeter 5-tuple flow records
with 80+ statistics, computed at the target host

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What a controller observes
PacketIn messages, FlowMod decisions,
per-entry counters of the rules its app installed

NOVELTY

Contributions: what is new and what is already known

Already known

Identifier columns are shortcut features (Engelen 2021)

Random splits leak on NIDS data (Flood 2024)

Control-plane telemetry datasets exist (LAN-SDN-NIDS)

New here

Controller-telemetry gap inside InSDN: 0 FlowMods match IPs or ports

Paired random vs host-held-out gap on InSDN and CICIDS2017, 10 seeds, CIs

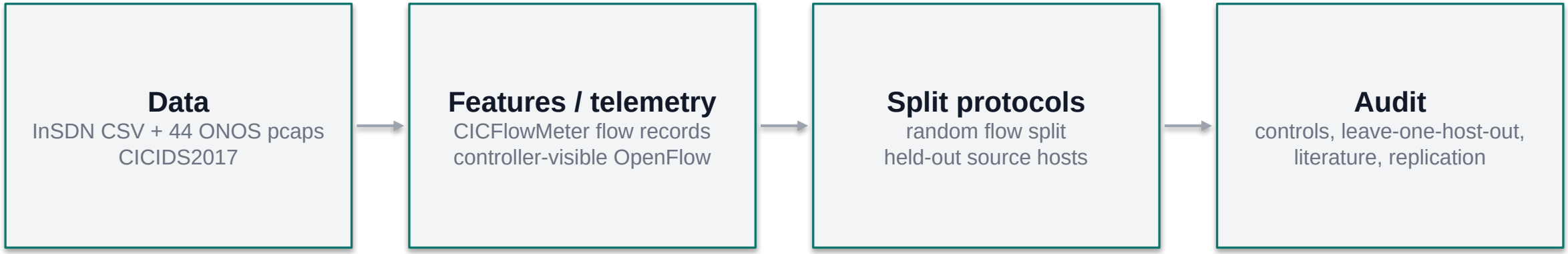
Host structure and leave-one-host-out detection

Ten published papers read in full; none host-grouped

Not claimed: a new detector, or results on a second SDN dataset or controller family.

ARCHITECTURE

Pipeline: data, controller view, split protocols, audit



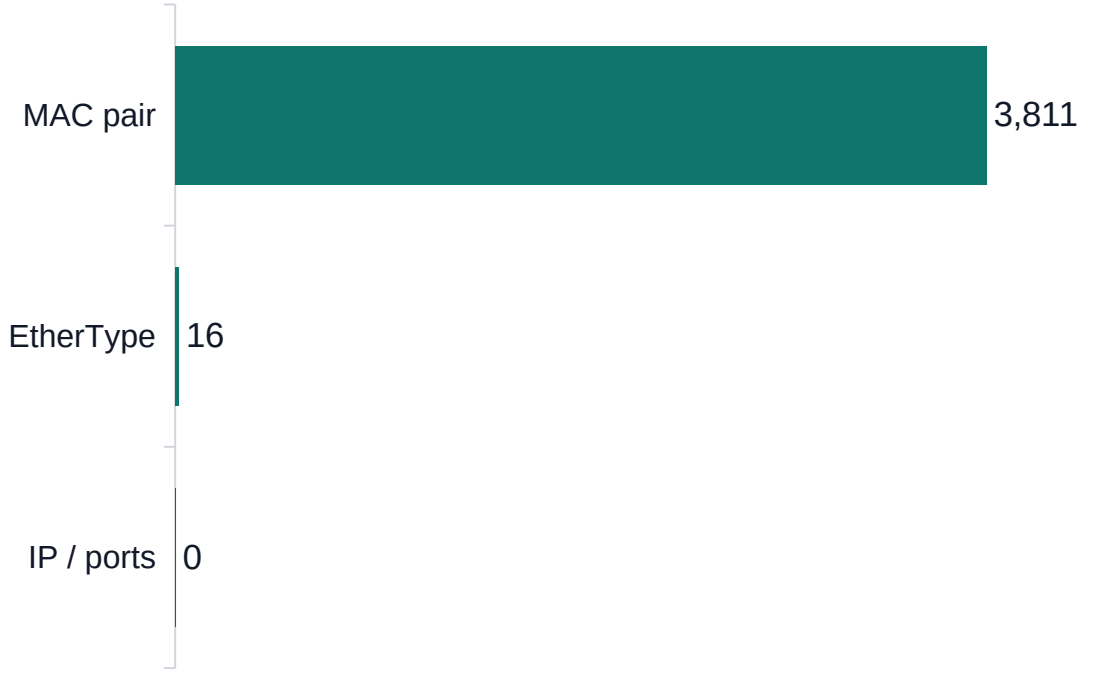
Every number comes from results/*.json, 10 seeds with 95% CIs; captures are SHA-256 hashed and the results rerun from one script.

OBSERVABILITY GAP

The controller never saw the labelled flows

0 of 3,827

FlowMods in the 44 captures match IP addresses or transport ports

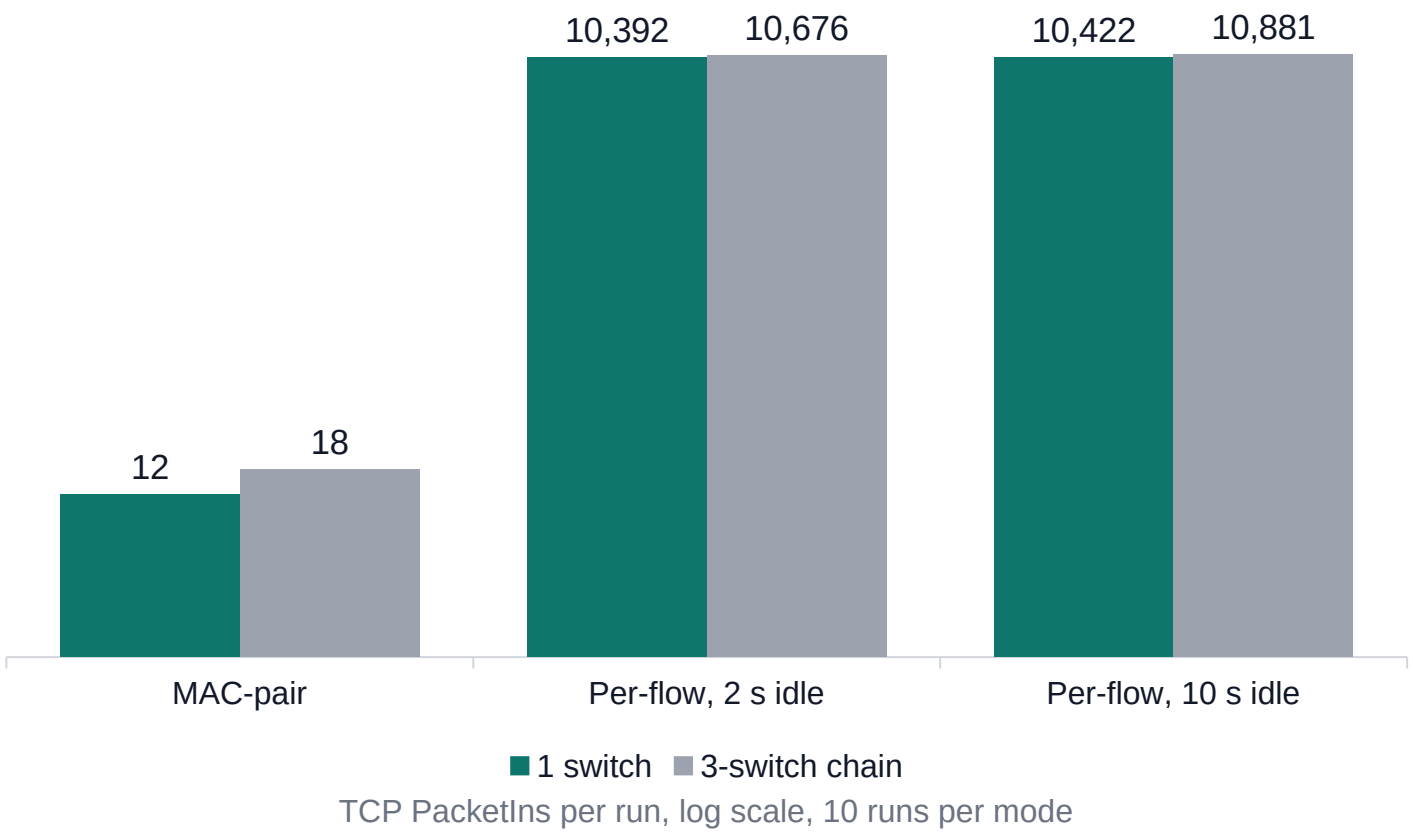


FlowMods by match structure

Ryu's stock learning switch agrees: 91 FlowMods in 5 captures, none with IP or transport fields. The dataset's 5-tuple flows are not counters a controller can read.

FORWARDING CASE STUDY

The forwarding app sets what the controller observes



TCP PacketIns per run, log scale, 10 runs per mode

847×

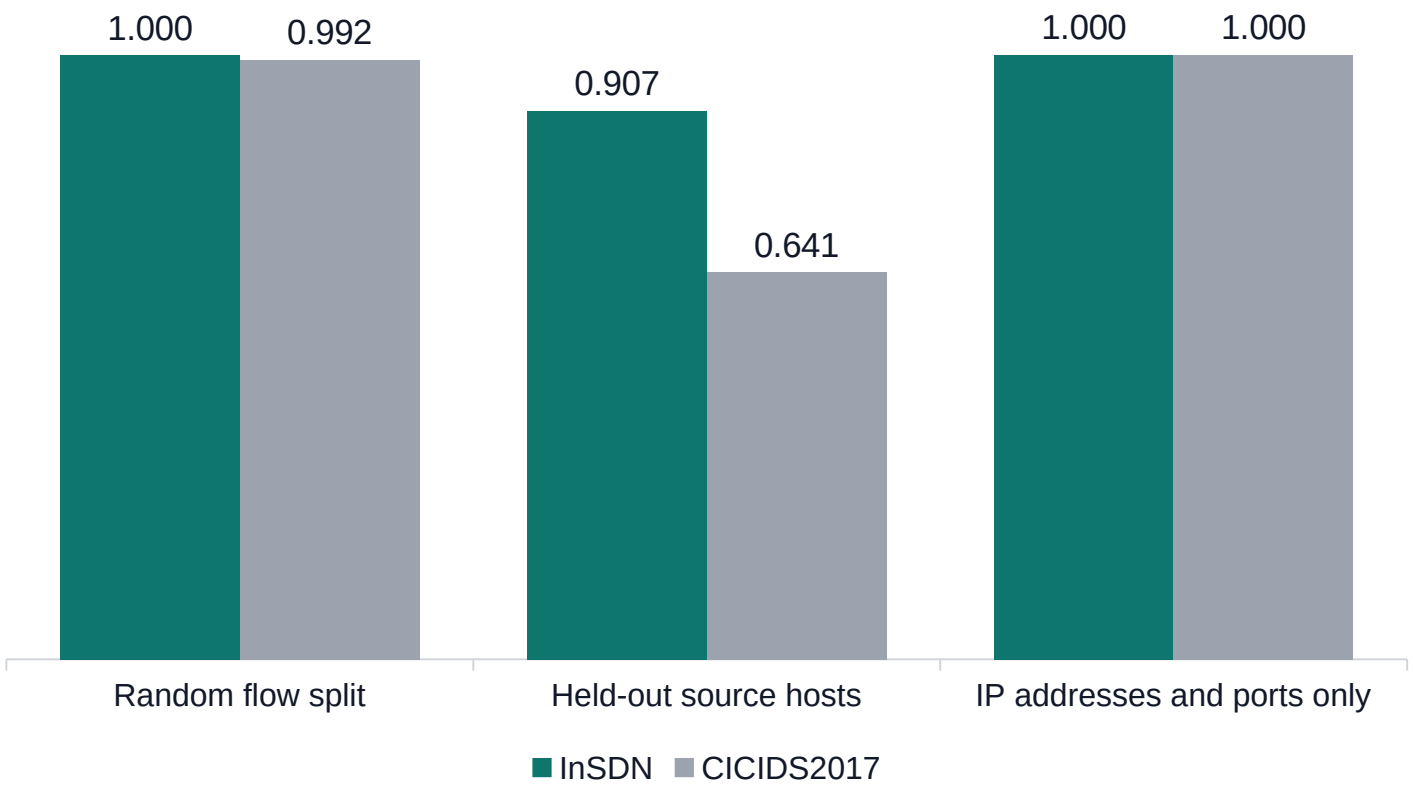
more PacketIns with per-flow rules on identical traffic

A 2 s idle timeout cuts the entries a 1 s poller sees from 153 to 37 per reply.

Our first app let ping/ARP rules shadow TCP and showed 4×. We report both.

SPLIT-PROTOCOL AUDIT

Random splits inflate scores: 9 points (InSDN), 35 (CICIDS2017)



Balanced accuracy, GPU XGBoost, 10 seeds, benign vs attack. 343,889 / 2,100,814 flows

49% / 20%

of random-split test rows are exact duplicates of a training row (InSDN / CICIDS2017)

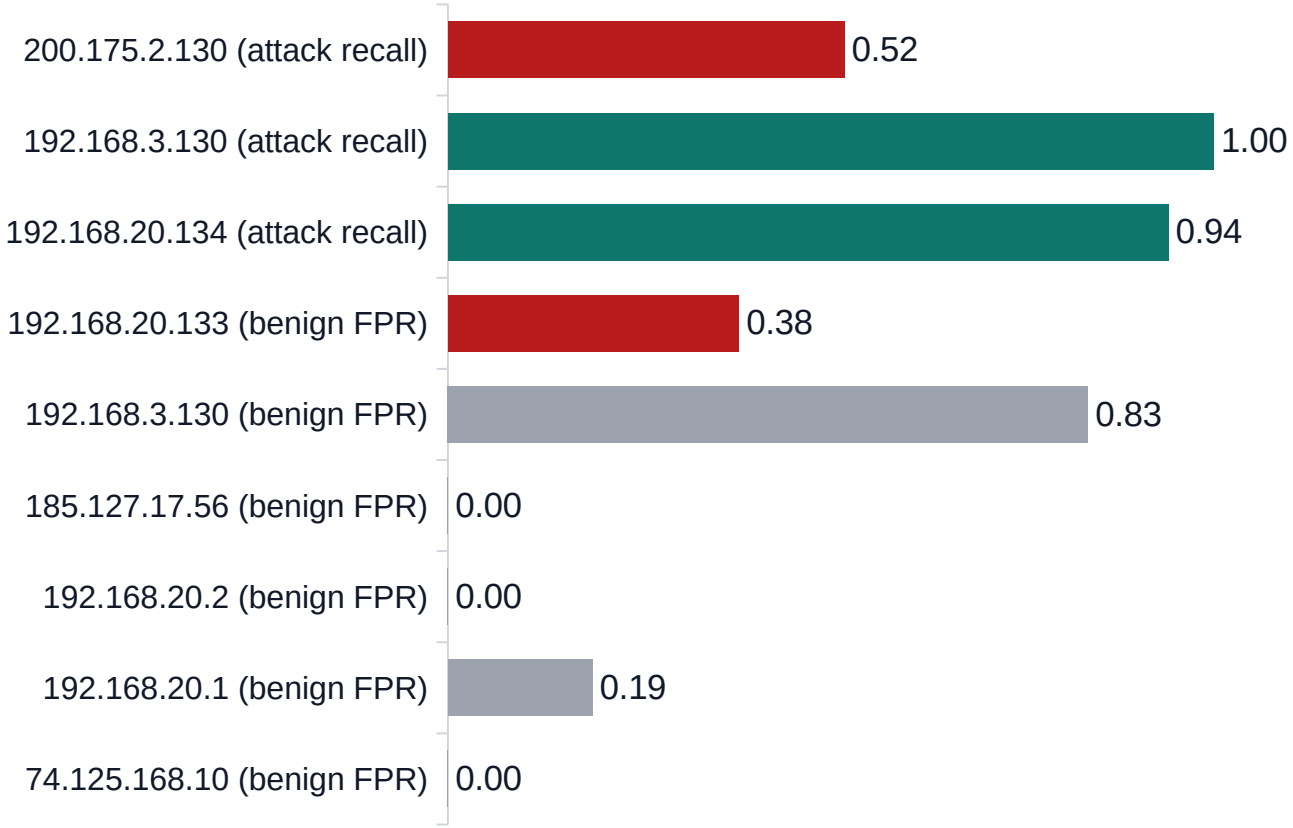
Held-out hosts: 0.907 ± 0.085 (InSDN, worst seed 0.66), 0.641 ± 0.056 (CICIDS2017).

Identifiers alone reach 1.000 on both, so a random split rewards memorising hosts.

Known already: Engelen 2021, Flood 2024. New: the paired gap with CIs on an SDN dataset.

Held-out attacker hosts are poorly detected, on both datasets

122,868 source IPs, 122,307 with a single flow; 8 hosts with 500 or more flows hold 215,404 of 343,889 rows.



Train on all other hosts, test on the held-out host. Green bars: attack recall; grey: benign false-positive rate; red: weak held-out host.

CICIDS2017

48 source IPs, 19 with a single flow; 15 hosts hold nearly all 2,100,814 rows.

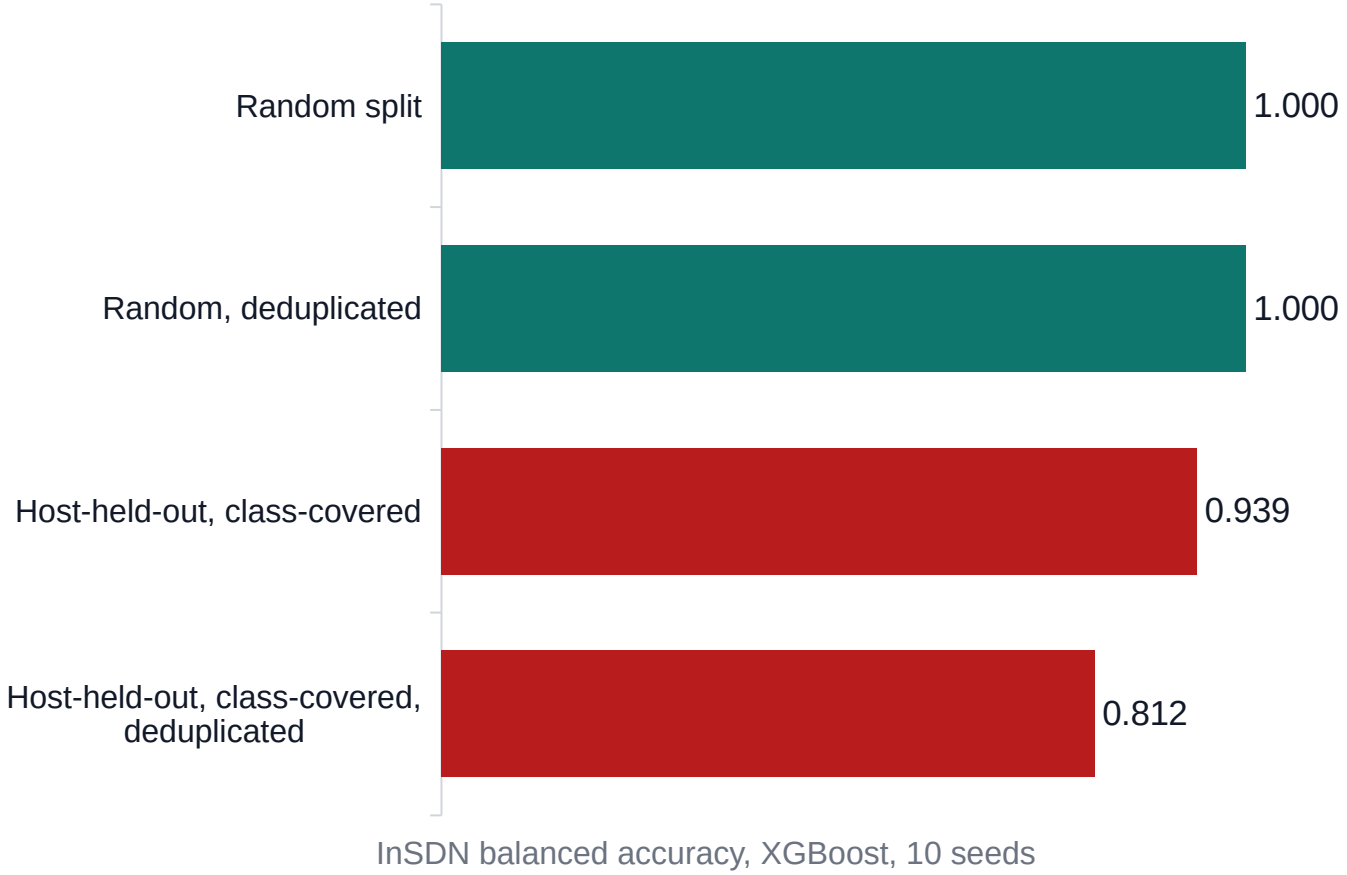
0.28

attack recall when the main attacker 172.16.0.1 is held out

0.54 of its benign flows are flagged. Training on all other hosts, XGBoost; identical across 10 model seeds.

CONTROLS

Duplicates and class absence do not explain the drop



InSDN balanced accuracy, XGBoost, 10 seeds

Removing exact duplicates (46% of rows) leaves the random split at 0.9998.

Keeping only splits where every attack class is in training still gives 0.939 (CI ±0.06).

Class-covered and deduplicated: wide CI, so we report it as a stress test.

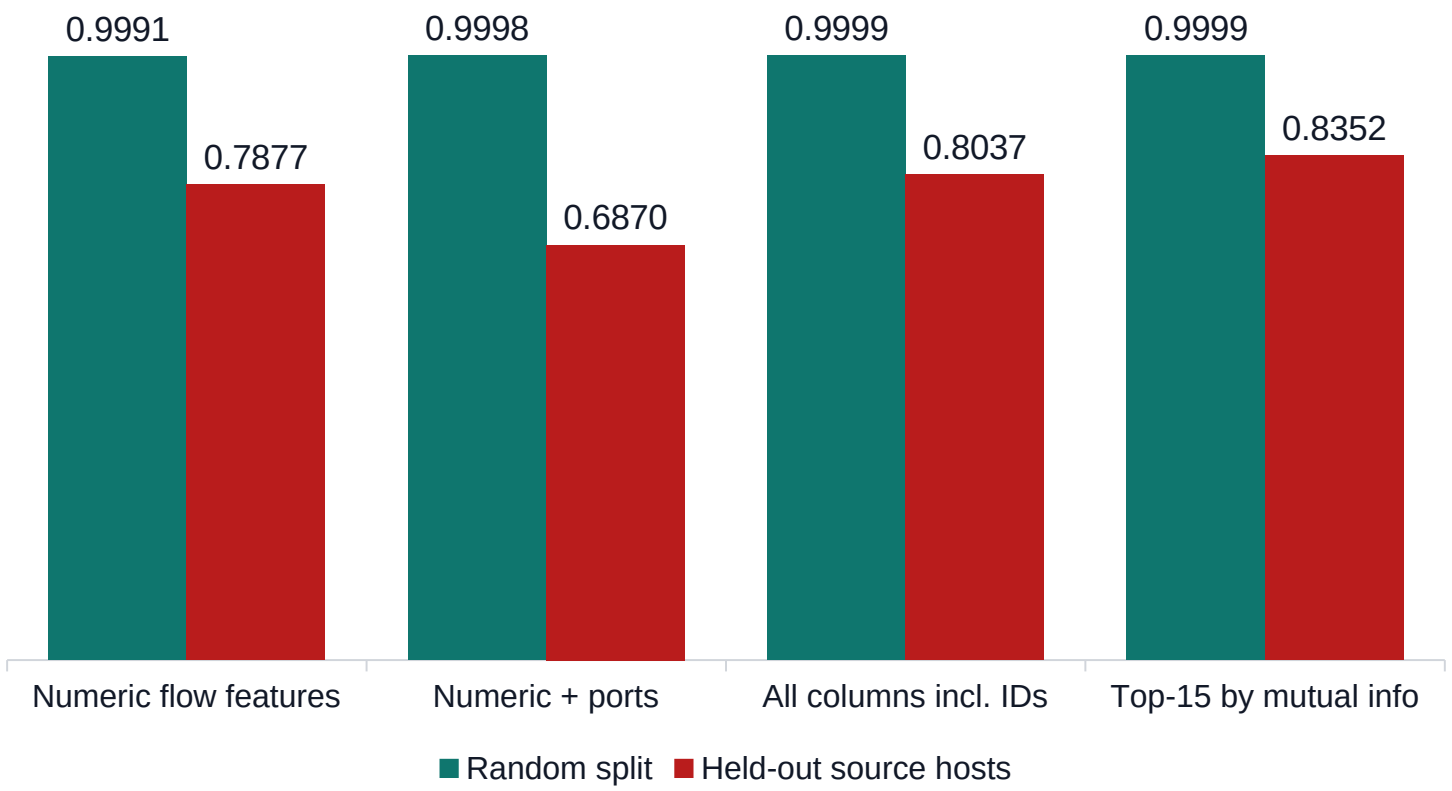
PROTOCOL AUDIT OF THE LITERATURE

Published papers rarely describe their split

Paper	Reported	Split	Grouped by host	IDs / ports as inputs
Towhid et al.	RF 98.84%	70:30, all flows	No	Not stated
Ataa et al. 2024	99.02%	Not described (shuffled)	Not stated	No (stated)
OptiXGB-IDS 2026	99.87%	80:20 random	No	Ports
PeerJ CS 2025	99.93%	train_test_split	No	Not stated
Sajid et al. 2025	99.98%	Not described	Not stated	Dst Port
SDNGuardStack 2026	99.98%	80/20	No	Flow ID, IPs, ports
Malik et al. 2026	81.60% (CV)	Stratified 5-fold	No	Not stated
LightIDS-SDN 2026	98.73%	Ratios 60:40 to 90:10	No	Not stated
SICA 2026	99.61%	70/20/10	No	No (stated)

Ten papers read in full; most do not describe their split; none reports a host-grouped one. Table: the read papers that use InSDN; numbers are the authors' own, not re-run.

Published feature recipes, two splits



8-class accuracy, GPU XGBoost, 80:20, 10 seeds, 343,889 flows

Random split: at least 0.999 for every recipe.

Held-out hosts: 0.69 to 0.84, with 95% CIs of ±0.18 to ±0.24.

Adding ports lowers the held-out-host score (0.79 to 0.69) while the random-split score rises.

Caveat: one classifier (XGBoost) and approximate recipes of the published feature choices, not the papers' models or code. Grouping by source host is a stress test, not a deployment estimate.

PARTIAL REPLICATION

SDNGuardStack fails on unseen hosts

0.9999 Random split, our approximation (paper reports 99.98%)

0.72 to 0.78 Source-IP grouped split, two training variants

±0.39 Largest 95% CI across seeds on the grouped split

Partial replication: 5 seeds, an approximation of the paper's model and features, not its code. Grouping by host is a stress test, not a deployment estimate.

HONEST SCOPE

Limitations we state ourselves

Controller study: one dataset (InSDN) and one recorded controller (ONOS); Ryu is checked only on our own benign traffic.

We generated no attack traffic; benign and attack are never interleaved in one session.

Labels cover only part of the controller-visible traffic, so session identity and unlisted attacks cannot be separated.

Literature check: ten papers read in full; reported numbers are the authors' own, not re-run.

The split audit covers two datasets, one model family (XGBoost) and binary labels; UKMDDoSDN has no benign class and was unusable.

Our literature search (October 2026) is evidence, not proof, that no one has measured this on an SDN dataset.

TAKEAWAYS

Conclusion: what the paper defends

A controller cannot read the dataset's 5-tuple flows: 0 of 3,827 FlowMods match IPs or ports.

Controller-side scores on InSDN are a validity threat, not detection accuracy.

Random splits inflate scores on InSDN and CICIDS2017; held-out hosts expose it.

Neither duplicates nor class absence alone explains the gap.

Report host-grouped splits with CIs; state the split.

We defend the measurement and the protocol, not a new detector.